Internship Progress Seminar

Yeshwantrao Chavan College of Engineering, Nagpur

Department of Electronics Engineering | Session: 2025–26

Final Progress Seminar

Presented By: Khushi Narendra Atara
Class Roll No: 15
Registration No: 22070640

Company Details

Internship Company

Cognizant Technology Solutions Pvt. Ltd

Owner / CEO

Ravi Kumar (CEO, Cognizant)

Complete Address

Cognizant Technology Solutions, Chennai / Nagpur, India

Year of Establishment

1994

Contact Person

Ashly M. Benny (Trainer / Supervisor)

Contact Email

ashly.bennym@cognizant.com

Stipend

As per Offer Letter

Type of Internship

Training Based Internship (GenC QEA Program)

Work Progress — January to February 2026

WORK ASSIGNED

GenC QEA Onboarding: Behavioural Bootcamp (Icebreaker Week) — Corporate Induction, Soft Skills Training, and Development Environment Setup.

WORK DONE

- Attended Corporate Induction, Talent Manager Connect, Leader Talks, and Alumni sessions.
- Understood GenC program structure, RAG Performance Health Status (PHS) system, and Flipped Classroom model.
- Set up development environment: Eclipse IDE 2023-12, Azul JDK 17; navigated Tekstac platform.
- Completed Behavioural Training (ILT) on communication, teamwork, and corporate etiquette.
- Began Core Java (Stage 1): OOP Concepts, Data Types, Constructors, Exception Handling, Collections Framework.
- Completed Java Code Challenges and ICT Assessment (Core Java Skill War); appeared for Stage 1 Mock Assessment.

Work Progress — February to March 2026

WORK ASSIGNED

Stage 1 Completion: SQL Language, Stage 1 Final Assessment. Stage 2 Start: Functional Testing, STLC, Agile/Scrum, JIRA Tool.

WORK DONE

- Completed SQL (DDL, DML, JOINs, Subqueries, Aggregate Functions); cleared SQL Code Challenge Assessments.
- Appeared for Stage 1 Final Assessment (Core Java + SQL) — 3 SBA questions, 3 hours.
- Learned SDLC/STLC phases, testing types (Black Box, White Box, Regression, Smoke, Sanity, E2E Testing).
- Studied Agile/Scrum: Sprints, User Stories, Sprint Planning, Review and Retrospective ceremonies.
- Hands-on with JIRA: Project setup, Backlog, Epics, Bug creation, writing Test Cases with Zephyr plugin.
- Appeared for Functional Testing ICT Assessment; initiated Stage 2 Integrated Project (Agile deliverables).

Work Progress — March to April 2026

WORK ASSIGNED

Selenium WebDriver Automation
- Selenium architecture, locators, Dynamic XPath
- WebDriver Actions, Waits (Implicit/Explicit/Fluent)
- Page Object Model (POM) + Apache POI
- Data-Driven Testing using Excel sheets

TestNG Framework
- TestNG annotations, assertions, DataProvider
- Parallel execution, XML/JSON data parsing
- Extent Reports & Listeners

Interim Evaluation (Days 44–45)

WORK DONE

- Completed Selenium WebDriver setup, advanced locators (CSS, XPath, Shadow DOM, Chained Locators).
- Built POM framework with Page Factory; completed Apache POI Excel Read/Write operations.
- Completed Code Challenge Assessments for WebDriver Basics, Dynamic XPath, POM & Apache POI.
- Completed TestNG: assertions, DataProviders, XML/JSON parsing, parallel execution, Extent Reports.
- Attempted Selenium Mock ICT (Day 42) and Selenium ICT Assessment (Day 43).
- Delivered Mini Project (individual) to BU SME for review.
- Appeared for Interim Project Evaluation + Interim Technical Evaluation via video interview (Days 44–45).
- Integrated Project deliverables: POM Framework, TestNG Suite, DataProviders, testng.xml submitted.

Work Progress — April to May 2026

WORK ASSIGNED

Automation Framework Design, DevOps (GIT/Maven/Jenkins/Grid), BDD Cucumber, API Automation (Postman & REST Assured), Docker & Containers, GenAI Code Assist, Hackathon Project, and Final Evaluation.

WORK DONE

- Automation Framework: Designed Hybrid Framework (POM + Data-Driven + Keyword-Driven), Log4j logging, Extent/Allure Reports.
- DevOps: Practiced GIT (clone, commit, push, merge), Maven project structure, Jenkins CI pipeline, Selenium Grid parallel execution.
- BDD Cucumber: Wrote Feature files in Gherkin (Given-When-Then), Step Definitions; integrated Cucumber with Selenium framework.
- API Automation: Tested APIs with Postman (GET/POST/PUT/DELETE); automated with REST Assured (status codes, JSON/XML, auth methods).
- Docker: Set up Dockerized Selenium Grid for consistent cross-environment test execution.
- GenAI Code Assist: Used GitHub Copilot for AI-driven BDD generation, dynamic locator creation, and automation code suggestions.
- Hackathon Project (Group): Delivered complete automation framework; Final Project Evaluation by BU SME (Days 63–65).
- Appeared for Final Technical Evaluation + Final Project Evaluation via video interview (11–13 May 2026). Program completed.

Hackathon Project — Problem Statement

PROBLEM STATEMENT: Identify Courses

Search and display all web development courses from coursera.org (or similar site) — filter for Beginners level, English language — and display the first two courses with Name, Total Learning Hours, and Rating.

DETAILED DESCRIPTION — 3 TASKS

Task 1: Search for Web Development courses (Beginners, English). Extract course name, total learning hours, and rating for first 2 courses.

Task 2: Look for Language Learning section. Extract all languages, different levels, with their total count and display them.

Task 3: Home Page → For Enterprise → Courses for Campus. Fill 'Ready to Transform' form with one invalid input (e.g. invalid email). Capture and display the error message.

Hackathon Project — Automation Scope & Tech Stack

KEY AUTOMATION SCOPE

- Handling different browser windows and search functionality
- Extract multiple dropdown list items and store in Java Collections
- Navigating back to home page (browser navigation)
- Filling web forms with multiple input field types
- Capture and validate warning/error messages
- Scrolling down in web pages using JavaScript Executor
- Applying Dynamic XPath for dynamic web elements
- Screenshot capture on test failure

TECHNOLOGY STACK

Language: Java (Core Java + OOP)

Automation Tool: Selenium WebDriver 4.15

Test Framework: TestNG with testng.xml

Design Pattern: Page Object Model (POM)

Data Handling: Apache POI (Excel)

BDD: Cucumber + Gherkin

Build Tool: Maven

CI/CD: Jenkins

Reporting: Extent Reports

IDE: Eclipse IDE 2023-12

Hackathon Project — Framework Architecture & Key Features

POM Framework Structure

```
src/main/java
  ├── pages/
  │   ├── HomePage.java
  │   ├── CoursePage.java
  │   └── FormPage.java
  └── utils/
      ├── ExcelUtils.java
      └── ScreenshotUtils.java

src/test/java
  ├── tests/
  └── runner/
```

Key Automation Features

- ✓ Multi-window handling
- ✓ Dynamic XPath locators
- ✓ Dropdown extraction to List
- ✓ Form submission & error capture
- ✓ Scroll via JS Executor
- ✓ Home page navigation
- ✓ Apache POI data-driven
- ✓ Extent Report generation
- ✓ Screenshot on failure
- ✓ Cucumber BDD scenarios

Test Execution Flow

1. Launch Browser (Chrome)
2. Navigate to coursera.org
3. Task 1: Search → Filter → Extract 2 courses
4. Task 2: Language Learning → Extract all levels + counts
5. Task 3: Enterprise → Campus form → Invalid email → Capture error
6. Generate Extent Report
7. Jenkins CI triggers on commit
8. Results published via Maven

Key Learnings & Program Completion

Stage 1 — Core Java & SQL

OOP Concepts, Collections, Exception Handling, JDBC, SQL DDL/DML/JOINs/Subqueries. Built solid foundation for enterprise Java development.

Stage 2 — Selenium & Testing

Selenium WebDriver, POM, Apache POI, TestNG, BDD Cucumber, REST Assured API automation, DevOps with Jenkins & Docker.

Project Skills

Designed a complete Hybrid Automation Framework. Applied Agile/Scrum via JIRA. Delivered Mini Project (individual) and Hackathon Project (group).

GenAI & Industry Tools

Used GitHub Copilot for AI-assisted automation. Learned CI/CD with Jenkins, GIT version control, Maven build lifecycle, and Docker containerization.

Thank You

Khushi Narendra Atara

GenC QEA — Quality Engineer with Selenium Automation (v3)

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